



# Lecture 9: Rocky Planets I Earth & Venus

## Introduction to Planetary Science

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# Learning Objectives

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By the end of this lecture, you will be able to:

- Summarise Earth's interior, surface, ocean, and climate as the reference planet
- Compare Venus to Earth across mass, radius, atmosphere, and rotation
- Explain why a saturated water atmosphere has a maximum thermal flux
- Derive the Simpson–Nakajima runaway greenhouse limit
- Describe the D/H smoking-gun evidence for past water on Venus
- Outline current and upcoming Venus missions and the questions they target

## Recap: Lectures 1–8

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- Solar system formed from a disk of gas and dust around 4.567 Ga
- Heat sources: accretion, radiogenic, tidal, residual primordial
- Differentiation produces a layered interior: crust, mantle, core
- Atmospheric escape, magnetospheres, surface processes
- Earth's interior:  $C/MR^2 = 0.331$ , liquid outer core, vigorous convection

**Today:** the two innermost rocky planets, very similar in bulk parameters, very different in evolution. **Why?**

# 1. Earth as Reference

Earth seen from ~ 6 billion km. Voyager 1 (1990). Credit: NASA/JPL-Caltech

## Earth at a Glance

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| Parameter         | Value                                     | Note                                       |
|-------------------|-------------------------------------------|--------------------------------------------|
| Mass              | $5.972 \times 10^{24} \text{ kg}$         | defines $M_{\oplus}$                       |
| Radius (mean)     | 6371 km                                   | defines $R_{\oplus}$                       |
| Mean density      | $5515 \text{ kg m}^{-3}$                  | uncompressed $\sim 4030 \text{ kg m}^{-3}$ |
| Semimajor axis    | 1.000 AU                                  | defines AU                                 |
| Bond albedo       | 0.30                                      | cloud + ice + land                         |
| $T_{\text{eq}}$   | 255 K                                     | equilibrium, no greenhouse                 |
| $T_{\text{surf}}$ | 288 K                                     | +33 K greenhouse warming                   |
| Rotation period   | 23.93 h                                   | spin axis tilt $23.4^{\circ}$              |
| Magnetic field    | dipolar, $\sim 25\text{--}65 \mu\text{T}$ | active dynamo                              |

Source: NASA Earth Fact Sheet (NSSDC)

# What Makes Earth Habitable?

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Four ingredients, all present, all stable for ~ 4 Gyr:

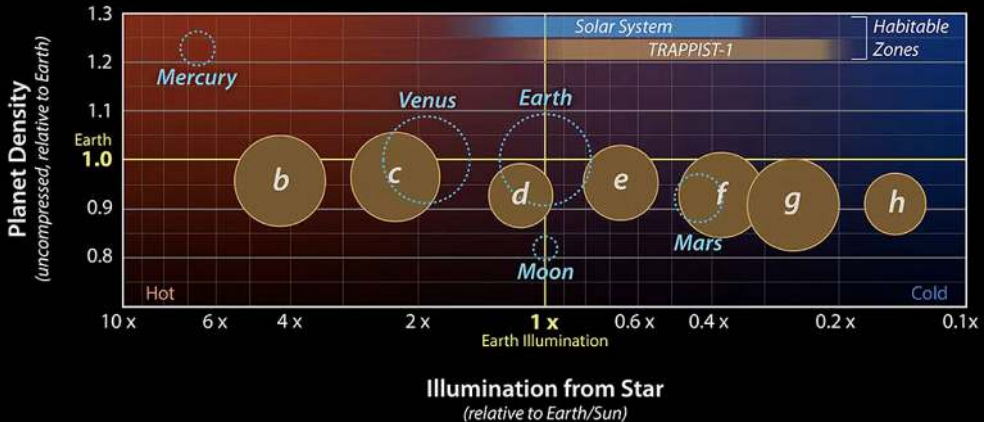
1. **Liquid water** on the surface, continuously
2. **Plate tectonics**: long-term carbon cycle, climate buffering
3. **Magnetic field**: shields atmosphere from solar wind erosion
4. **Stable orbit**: low eccentricity, large stabilising Moon

None of these is unique to Earth in principle.

All four together, on the same planet, at the same time, may be.

The puzzle is not that Earth is special. It is to identify which features are necessary, which are sufficient, and which are accidental.

# TRAPPIST-1/Solar System Comparison



Credit: NASA

## Why Compare to Other Systems?

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Earth is one of ~ 5800 confirmed planets. The natural comparison set:

- Inner solar system: 4 rocky planets, 0.4–1.5 AU, around a G-type star
- TRAPPIST-1: 7 Earth-sized planets, 0.011–0.062 AU, around a late M dwarf
- Both systems span the inner edge of the habitable zone
- Comparable orbital architecture, drastically different host stars

Earth is the calibration point. Whether it is the typical or atypical outcome of rocky-planet evolution is exactly what we are trying to find out.



# Plate Tectonics: The Engine

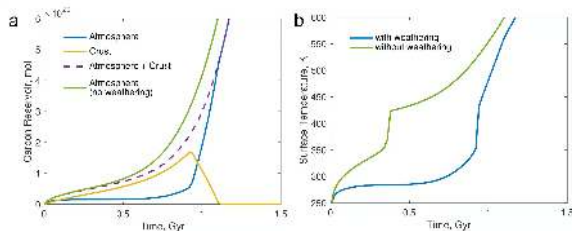
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- Earth's lithosphere is broken into ~ 15 major plates moving at  $\text{cm yr}^{-1}$
- Three boundary types:
  - **Divergent** (mid-ocean ridges): new crust forms
  - **Convergent** (subduction zones, mountain belts): crust destroyed or thickened
  - **Transform** (e.g. San Andreas): lateral slip
- Driven by mantle convection, slab pull, ridge push
- Recycles surface volatiles into the mantle on ~ 100 Myr timescales

**No other body in the solar system has active plate tectonics today.**

# Earth's Long-Term Interior Evolution

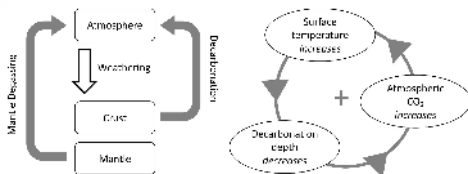
A coupled-model reference evolution over 4.5 Gyr:



Credit: Höning et al. (2021)

- Mantle temperature drops by  $\sim 600$  K
- Plate speed declines as the mantle cools
- Continental coverage and atmospheric  $\text{CO}_2$  track the slowing tectonic engine
- Sets the baseline for "what if Venus had been like this"

# The Long-Term Carbon Cycle



Credit: Höning et al. (2021)

Three coupled fluxes set atmospheric CO<sub>2</sub>:

- **Volcanic outgassing:** mantle to atmosphere
- **Silicate weathering:**  
 $\text{CaSiO}_3 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{SiO}_2$
- **Carbonate subduction + decarbonation:** crust to mantle

Weathering rate increases with  $T_s$ : a negative feedback that buffers Earth's climate.

# Earth's Magnetic Field

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- Generated by convection in the liquid outer core (Lecture 4)
- Mostly dipolar, surface intensity 25–65  $\mu\text{T}$
- Magnetopause stand-off:  $\sim 10 R_{\oplus}$  on the dayside
- Reverses polarity on  $10^5$ – $10^6$  yr timescales (paleomagnetism)
- Active for at least 3.45 Ga (Tarduno et al. 2010)

**Key consequence:** deflects solar wind, suppresses ion-pickup escape of  $\text{O}^+$ ,  $\text{N}_2^+$ ,  $\text{H}^+$ .

Comparison to unmagnetised Mars (Lecture 10) shows the magnitude of this effect.

# Hydrosphere and Cryosphere

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## Liquid water reservoirs:

- Oceans:  $1.34 \times 10^9 \text{ km}^3$
- Groundwater:  $\sim 2.3 \times 10^7 \text{ km}^3$
- Lakes, rivers, soil moisture: small

## Ice reservoirs:

- Antarctica:  $26.5 \times 10^6 \text{ km}^3$
- Greenland:  $2.85 \times 10^6 \text{ km}^3$
- Mountain glaciers, permafrost: smaller

- Total surface water  $\approx 2700 \text{ m}$  global equivalent layer (GEL)
- Ocean is the largest pool, but the deep mantle holds  $\geq 1\text{--}2$  ocean masses
- Ice albedo provides a strong positive feedback in cold climate excursions

## Climate Stability and the Faint Young Sun

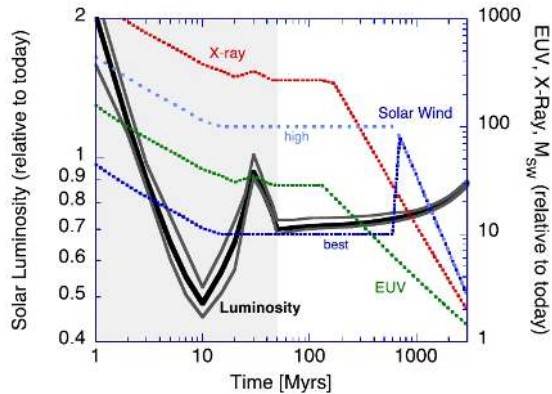
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Earth's climate has been stable enough for liquid water for ~ 4 Gyr, but not constant.

- Solar luminosity at 4 Ga: ~ 75% of today
- Without a strong greenhouse, Earth would have been frozen
- Geological evidence (sedimentary rocks, evaporites): liquid water at  $\geq 3.8$  Ga
- Compensating mechanism: higher  $\text{CO}_2 + \text{CH}_4$ , dimmer surface, weaker weathering

The Faint Young Sun problem is solved by the carbon cycle's negative feedback, not by accidents of initial conditions.

# Solar Luminosity Through Time

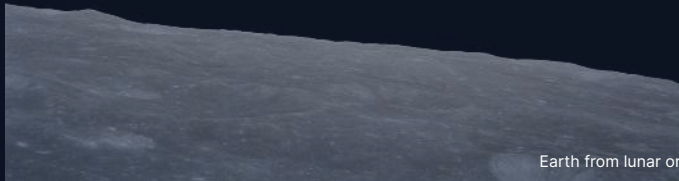


Credit: Zahnle et al. (2007)

- Sun has brightened ~ 30% since formation
- Habitable-zone inner edge has migrated outward
- Venus inside it for at least the last ~ 1 Gyr
- The boundary swept past Venus' orbit in the past

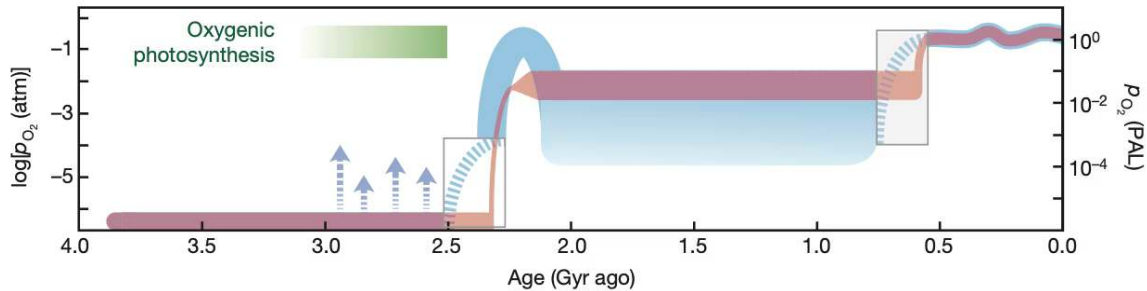
Question for later: was Venus on the wet side when the boundary swept past?

## 2. Earth's Atmosphere and Biosphere



Earth from lunar orbit, Apollo 8 (1968). Credit: NASA





## Reading the Oxygen History

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Two great rises in atmospheric O<sub>2</sub>:

- **Great Oxidation Event** (GOE) at ~ 2.4 Ga: trace → ~ 1% of present atmospheric level (PAL)
- Long Mesoproterozoic plateau at 1%–10% PAL
- **Neoproterozoic Oxygenation Event** (NOE) at ~ 0.6 Ga: → near modern levels
- Phanerozoic fluctuations of ~ 15–35%

**Drivers:** cyanobacterial photosynthesis + organic carbon burial + decreasing reductant flux from the mantle.

Co-evolution of biosphere, atmosphere, and tectonics, across more than 2 Gyr.

# Snowball Earth Episodes

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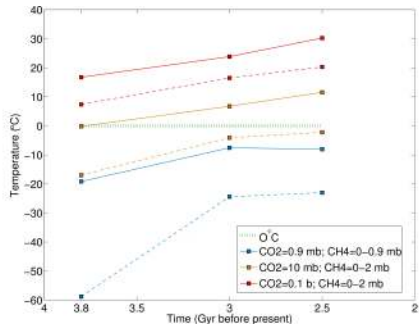
At least two global glaciation events in the Neoproterozoic:

- **Sturtian glaciation** (~ 717–660 Ma)
- **Marinoan glaciation** (~ 650–635 Ma)
- Glacial deposits at near-equatorial paleolatitudes
- Cap carbonates immediately above glacial diamictites

**Mechanism:** ice albedo runaway  $\Rightarrow$  frozen oceans  $\Rightarrow$  weathering halts  $\Rightarrow$  volcanic  $\text{CO}_2$  accumulates  $\Rightarrow$  extreme greenhouse triggers deglaciation.

Source: Hoffman et al. (1998, 2017)

# The Archean Climate Problem

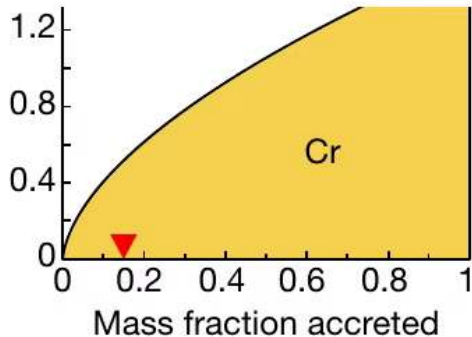


Credit: Charnay et al. (2013)

- Solar luminosity at 3 Ga: ~ 80% today
- Yet evidence for liquid water + biosphere
- Required greenhouse:  $CO_2$ ,  $CH_4$ , possibly  $N_2$  pressure broadening
- Constraints from paleosols, banded iron formations

Archean Earth was kept warm by an order of magnitude more greenhouse gas than today, not by a closer Sun.

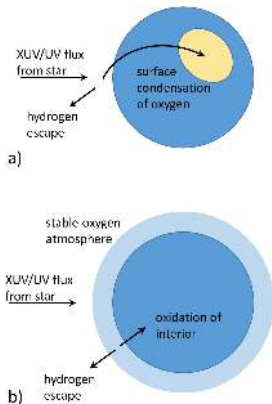
## Earth's Bulk Composition: Late Accretion



Credit: Dauphas (2017)

- Earth's bulk composition matches enstatite chondrites in  $\Delta^{17}\text{O}$ ,  $\epsilon^{50}\text{Ti}$ ,  $\epsilon^{54}\text{Cr}$
- Implies most of Earth's mass came from inner-solar-system material
- Late accretion (< 1% by mass) added water + highly siderophiles
- Earth is not a generic "average chondrite", it has a specific isotopic signature

# Abiotic Pathways to $O_2$

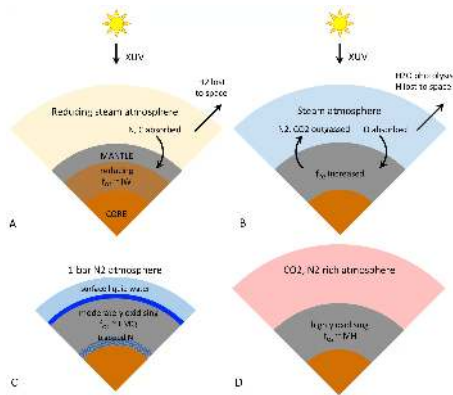


Credit: Wordsworth & Pierrehumbert (2014)

For exoplanet biosignatures: not all  $O_2$  implies life.

- $H_2O$  photolysis + H escape leaves abiotic  $O_2$
- Especially at runaway-greenhouse conditions
- Or in  $CO_2$ -rich atmospheres where  $CO + O_2$  recombination is slow
- Earth's  $O_2$  context-dependent biosignature

# The $N_2$ Redox Pump



Credit: Wordsworth (2016)

- Atmospheric  $N_2$  inventory is set by the redox state of the surface and atmosphere
- Biological denitrification + nitrification couple  $N_2$  to  $O_2$  history
- Solar wind nitrogen escape over Gyr can be substantial without a magnetic field
- Nitrogen tells us about both biology and magnetic shielding

## The Biosphere as Geological Force

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Life is not a passive passenger of Earth's climate; it is a geological agent.

- **Banded iron formations** (~ 3–1.8 Ga): biological  $O_2$  + dissolved  $Fe^{2+}$
- **Continental crust composition**: shifted by oxidative weathering
- **Ocean redox state**: from anoxic to oxic, Snowball linked to NOE
- **Atmospheric  $N_2$  inventory**: biological denitrification + nitrification

Earth's atmosphere is in extreme **thermodynamic disequilibrium** ( $O_2$  +  $CH_4$  coexist) maintained continuously by biology.



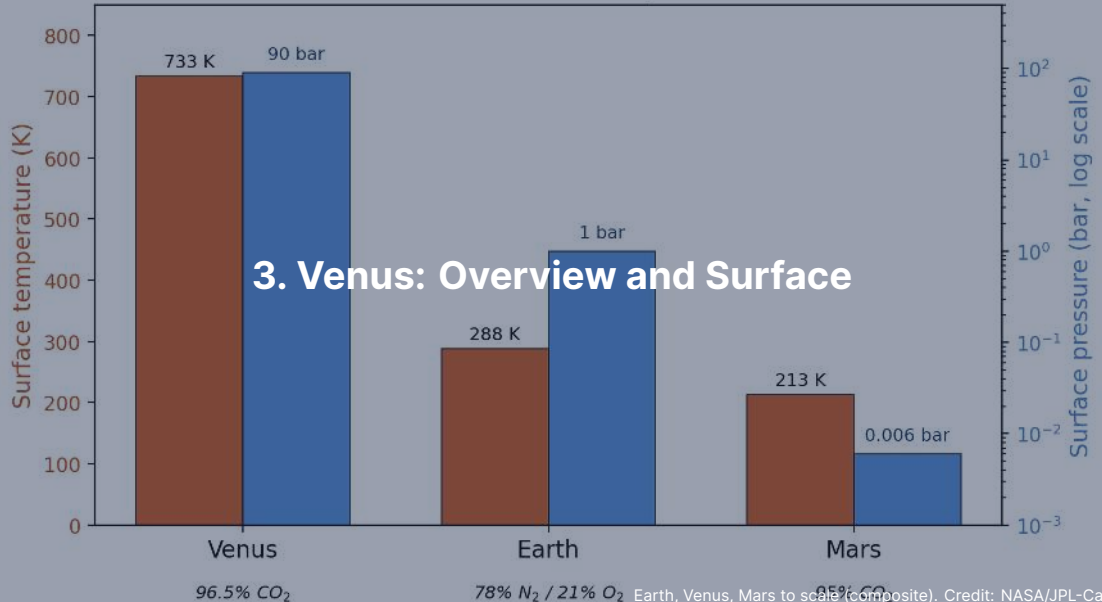
## Earth as Reference: Summary

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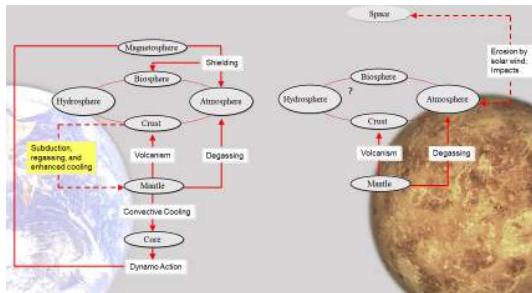
- Differentiated, magnetised, tectonically active rocky planet at 1 AU
- Liquid surface water, continuously, for ~ 4 Gyr
- Long-term carbon cycle buffers climate against luminosity changes
- Biosphere: a major geochemical reservoir, sink/source for many volatiles
- Plate tectonics + magnetic field + biosphere + ocean: the four-way coincidence

Now to the planet that started as Earth's apparent twin and ended as something completely different.

# Surface conditions and dominant atmospheric species for the three terrestrial planets



# Earth and Venus as Coupled Systems



Credit: Gillmann et al. (2022)

Both are coupled systems of:

- Interior + tectonics
- Magnetosphere or its absence
- Atmosphere + clouds
- Surface + (possible) hydrosphere

The differences between them are emergent properties of which feedback loops close.

## Venus and Earth: Twin and Not

| Parameter                           | Earth                   | Venus                           |
|-------------------------------------|-------------------------|---------------------------------|
| Mass ( $M_{\oplus}$ )               | 1.000                   | 0.815                           |
| Radius ( $R_{\oplus}$ )             | 1.000                   | 0.949                           |
| Mean density ( $\text{kg m}^{-3}$ ) | 5515                    | 5243                            |
| Semimajor axis (AU)                 | 1.000                   | 0.723                           |
| Stellar flux ( $S_{\oplus}$ )       | 1.00                    | 1.91                            |
| Bond albedo                         | 0.30                    | 0.77                            |
| $T_{\text{surf}}$ (K)               | 288                     | 737                             |
| Surface pressure (bar)              | 1.0                     | 92                              |
| Atmosphere                          | $\text{N}_2/\text{O}_2$ | $\text{CO}_2/\text{N}_2$        |
| Rotation period (d)                 | +1.0                    | -243 (retrograde)               |
| Magnetic field                      | $\sim 50 \mu\text{T}$   | $< 10^{-5} \times \text{Earth}$ |

Source: NASA Venus Fact Sheet (NSSDC)

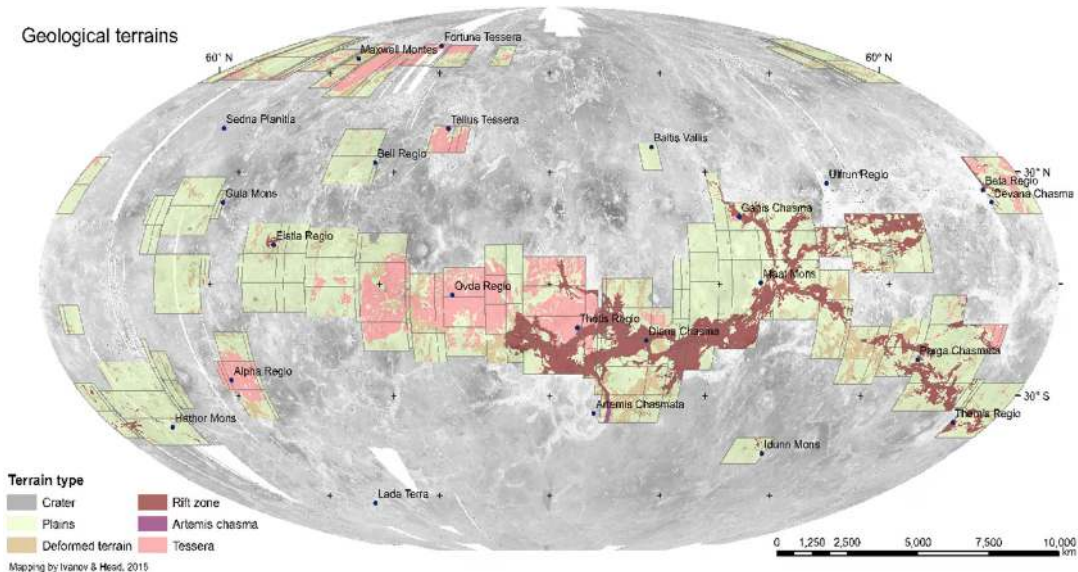
## Venus: A Long Mission History

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- **Mariner 2** (1962): first successful planetary flyby; thermal  $T_{\text{surf}} \sim 700 \text{ K}$
- **Venera series** (1961–1984, USSR): 13 atmospheric probes, 8 surface landers
  - Venera 4: first *in situ* atmospheric composition
  - Venera 9: first surface images (1975)
  - Venera 13/14: surface composition (XRF), basaltic crust
- **Magellan** (1989–1994, NASA): radar mapping at  $\sim 100 \text{ m}$  resolution, nearly global
- **Venus Express** (2005–2014, ESA): UV/IR atmospheric monitoring
- **Akatsuki** (2015–present, JAXA): atmospheric dynamics, super-rotation

No surface lander has survived more than  $\sim 2$  hours: deep-ocean-trench pressures at red heat.

## Geological terrains



## Reading the Venus Surface Map

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Magellan radar identifies several major terrain types:

- **Plains** (~ 80% of surface): basaltic, low elevation
- **Tessera**: highly deformed highlands, possibly oldest crust
- **Coronae**: circular tectonic-volcanic features
- **Rift zones** (e.g. Beta Regio)
- **Volcanic edifices** (> 85,000 catalogued)

Key observation: **no spatial pattern of crustal age**. No oceanic-style mid-ocean ridges, no clear subduction belts.

## Crater Statistics: A Single Surface Age

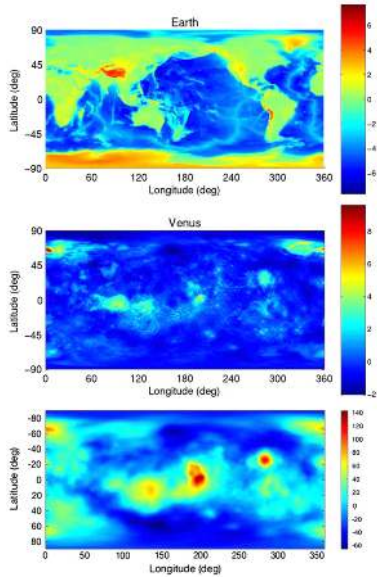
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Venus crater statistics from Magellan:

- ~ 1000 identified impact craters,  $\geq 2$  km diameter
- Distribution is **statistically random**, no major terrain younger than another
- Mean crater density implies a global surface age of ~ 300–1000 Ma
- No areas of clearly archaic ( $> 1$  Ga) terrain
- No active subduction belts (no oceanic-style recycling)

Venus' surface was either resurfaced globally and recently (~ 750 Ma catastrophic-resurfacing model) or is being resurfaced steadily (equilibrium model). Crater statistics alone cannot decide between them.





Earth topography, Venus topography, Venus geoid. Smrekar, Davaille & Sotin (2018)

## Why the Topography Differs

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- Earth: **bimodal hypsometry** (oceans ~ -4 km vs. continents ~ +1 km)
- Venus: **unimodal hypsometry**, single dominant elevation
- Earth's bimodality is a direct signature of plate tectonics + isostasy
- Venus has highlands (Aphrodite, Ishtar) but no continents in the Earth sense
- Geoid/topography ratio: deep compensation depths on Venus

Hypsometry alone is enough to rule out Earth-style plate tectonics on Venus.

## 4. Venus Interior and Tectonic Regime



## Venus' Interior: Inferred, Not Measured

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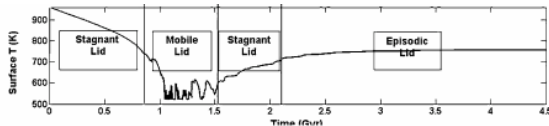
No seismometer has ever operated on Venus.

Interior structure is inferred from bulk parameters and Earth-analog models.

- Mean density  $5243 \text{ kg m}^{-3}$ ; uncompressed  $\sim 4060 \text{ kg m}^{-3}$
- $C/MR^2 \approx 0.33$  (assumed Earth-like, not measured)
- Implied core radius:  $R_c \approx 3000\text{--}3500 \text{ km}$  ( $\sim 0.55 R_V$ )
- Mantle composition: probably similar to Earth's
- Core state: possibly partially molten, but no strong evidence either way

**Key observational gap:** no  $C/MR^2$ , no seismicity, no high-resolution gravity.

# Venus Tectonic Evolution Scenarios



Credit: Smrekar, Davaille & Sotin (2018)

Two leading models:

- **Steady stagnant lid:** localised plumes, no global recycling
- **Episodic lid overturn:** long stagnant intervals + brief catastrophic mobilisation

Either way, no Earth-style plate tectonics today.

# The Missing Magnetic Field

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- Internal magnetic field  $< 10^{-5}$  that of Earth (Pioneer Venus, 1978)
- Possible reasons:
  1. Core not convecting (insufficient heat flux)
  2. Core fully solid (already crystallised)
  3. Core fully liquid but subadiabatic
  4. Slow rotation reduces dynamo efficiency
- Most likely: **stagnant lid** suppresses core heat loss  $\Rightarrow$  no convection in core

Without a magnetosphere, the upper atmosphere is directly exposed to the solar wind.

This is a major channel for Venus' water loss.

## Why Stagnant Lid?

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Earth-style plate tectonics needs:

1. A weak, brittle, water-saturated lithosphere
2. Negative thermal buoyancy at subduction zones
3. Continuous mantle return flow

Venus likely fails on point 1: hot, dry surface  $\Rightarrow$  strong, ductile lithosphere.

**Coupled-feedback view:** no liquid water  $\Rightarrow$  dry strong lithosphere  $\Rightarrow$  no plate motion  $\Rightarrow$  no carbon-cycle return  $\Rightarrow$  no water (one-way reinforcing loop).

Source: Smrekar et al. (2018); Gillmann et al. (2022)

## Is Venus Volcanically Active Today?

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Three independent pieces of evidence for recent or ongoing volcanism:

- **Magellan radar reanalysis** (Herrick & Hensley 2023): a  $\sim 2 \text{ km}^2$  vent on Maat Mons changed shape between 1991 and 1992
- **Venus Express thermal emissivity**: anomalously high emission from Idunn Mons, consistent with  $< 250 \text{ kyr}$  lava flows
- **Atmospheric  $\text{SO}_2$  variability**: order-of-magnitude changes on  $\sim$ decadal timescales, consistent with episodic outgassing

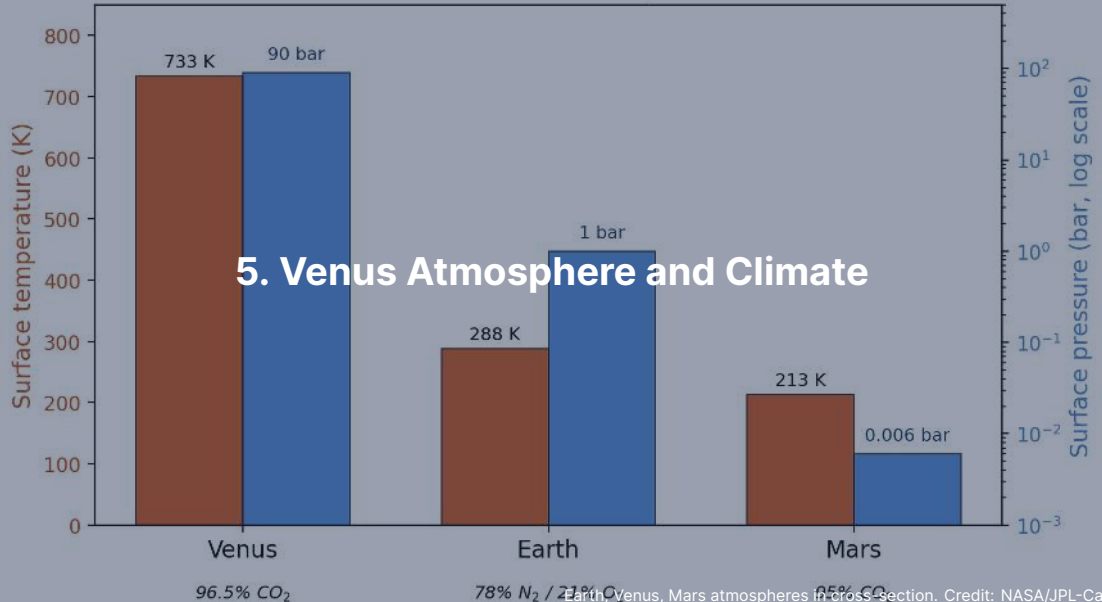
Venus is almost certainly volcanically active today. The questions are: at what rate, and from how many vents?



**Break**



## Surface conditions and dominant atmospheric species for the three terrestrial planets



## Venus Atmosphere: Structure

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- Surface: 737 K, 92 bar, 96.5% CO<sub>2</sub>, 3.5% N<sub>2</sub>
- Sub-cloud haze: 30–48 km
- **Main cloud deck:** 48–70 km, H<sub>2</sub>SO<sub>4</sub> droplets, ~ 75% wt
- Mesosphere + thermosphere above, with a hot dayside / cold nightside contrast
- No tropopause cold trap of the Earth kind
- Lower atmosphere: dry adiabat,  $\Gamma \approx 8 \text{ K km}^{-1}$

Total atmospheric mass: ~ 90× Earth's; almost entirely CO<sub>2</sub>.

# Atmospheric Super-Rotation

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Venus' atmosphere rotates much faster than the planet itself.

- Solid body: 243 days/rotation
- Cloud-top winds:  $\sim 100 \text{ m s}^{-1}$ , circling the planet in  $\sim 4$  days
- Angular velocity ratio:  $\sim 60\times$  the surface
- Driven by: thermal tides + large-scale waves transporting angular momentum
- Persistent for the full mission record ( $\sim 50$  years)

Super-rotation is the textbook example of an atmospheric regime with no Earth analog and no easy first-principles derivation.

## Venus' Greenhouse: $T_{\text{surf}}$ from $T_{\text{eq}}$

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Equilibrium temperature with  $A = 0.77$ :

$$T_{\text{eq}} = T_{\odot} \sqrt{R_{\odot} / (2a_V)} (1 - A)^{1/4} \approx 230 \text{ K}$$

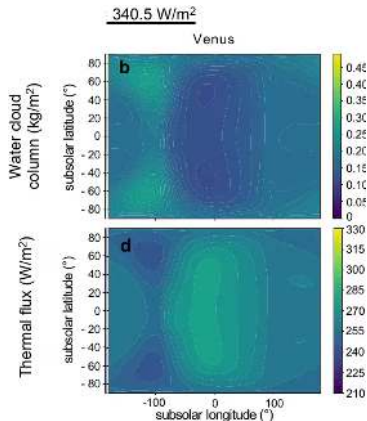
Actual surface temperature: 737 K.

Greenhouse effect:  $\Delta T = +500 \text{ K}$ .

- Earth:  $\Delta T = +33 \text{ K}$
- Venus:  $\Delta T = +500 \text{ K}$

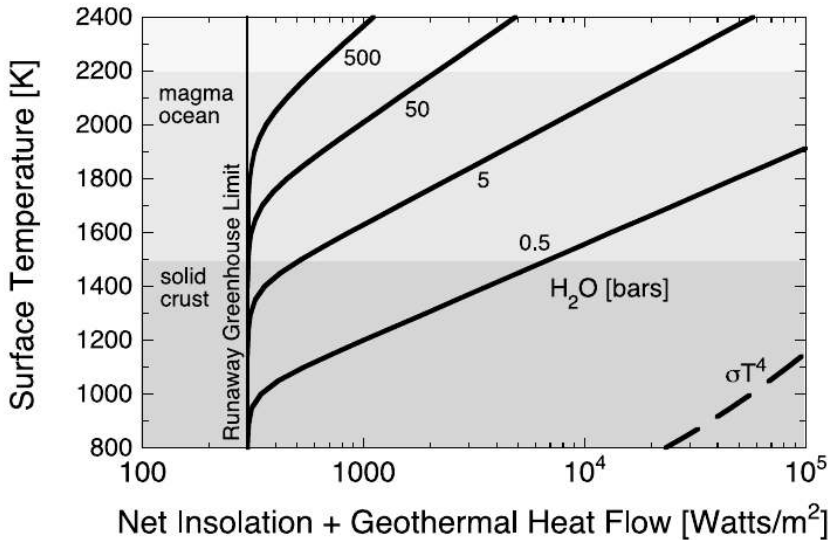
Venus is hot not because it is closer to the Sun (it absorbs less stellar flux than Earth!) but because of its  $\text{CO}_2$ -dominated, 92-bar atmosphere.

# Water Clouds and Emission Spectra



Credit: Turbet et al. (2021)

- Modelled emission spectra of dry-Venus vs. wet-Venus atmospheres
- Water-cloud signatures distinct in mid-IR
- JWST + future thermal-IR missions can distinguish wet/dry exo-Venus analogs
- Bridges Venus' history to exoplanet observability



## Reading the Runaway Threshold Figure

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- Horizontal axis: net heat flux that must be radiated to space
- Vertical axis: surface temperature in steady state
- For low flux: cool, stable solution
- Above a critical threshold ( $\sim 280 \text{ W m}^{-2}$ ): no steady state with surface liquid water
- Surface temperature **runs away** until water is fully vaporised
- Vertical line on the right: insolation threshold at which runaway is triggered

Maximum OLR sets the threshold. We derive it next, on the blackboard.



## 6. Blackboard Derivation: Simpson–Nakajima Limit



## Setup: Saturated Atmosphere, Greybody

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**Goal:** show that a water-saturated atmosphere has a maximum OLR independent of surface temperature.

### Assumptions:

- Atmosphere saturated in water vapour everywhere
- Single grey IR opacity  $\kappa$ , units  $\text{m}^2 \text{kg}^{-1}$
- Hydrostatic equilibrium, surface gravity  $g$
- Photosphere: where IR optical depth  $\tau \sim 1$

Clausius–Clapeyron sets the water vapour partial pressure:

$$\frac{dp_{\text{sat}}}{dT} = \frac{L p_{\text{sat}}}{R_v T^2}$$

with  $L \approx 2.5 \times 10^6 \text{ J kg}^{-1}$ ,  $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$ .

## Step 1: The Photosphere Column Density

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An IR optical depth of unity is reached at column density:

$$N_{\text{phot}} \sim \frac{1}{\kappa}$$

Hydrostatic equilibrium converts column density to pressure:

$$p_{\text{phot}} = g N_{\text{phot}} = \frac{g}{\kappa}$$

In a water-saturated atmosphere, the local water-vapour pressure equals saturation:

$$p_{\text{sat}}(T_{\text{phot}}) = \frac{g}{\kappa}$$

This single equation pins the photosphere temperature.

## Step 2: Photosphere Temperature

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Integrate Clausius–Clapeyron:

$$p_{\text{sat}}(T) \approx p_{\text{ref}} \exp \left[ -\frac{L}{R_v} \left( \frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Set  $p_{\text{sat}} = g/\kappa$  and solve for  $T$ :

$$T_{\text{phot}} \approx \frac{L/R_v}{\ln(p_{\text{ref}} \kappa/g) + L/(R_v T_{\text{ref}})}$$

Plug in  $g = 10 \text{ m s}^{-2}$ ,  $\kappa = 5 \times 10^{-2} \text{ m}^2 \text{ kg}^{-1}$ :

$$T_{\text{phot}} \approx 260 \text{ K}$$

**No surface temperature appears anywhere on the right-hand side.**

### Step 3: The OLR Asymptote

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OLR is approximately blackbody emission at  $T_{\text{phot}}$ :

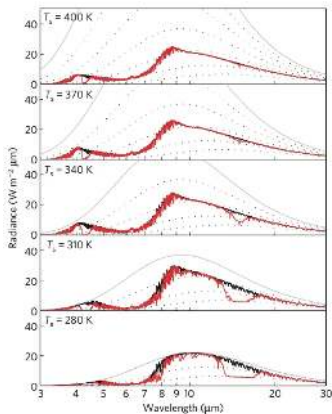
$$F_{\text{OLR}} \approx \sigma T_{\text{phot}}^4$$

$$F_{\text{OLR}}^{\text{max}} \approx 5.67 \times 10^{-8} \times (260)^4 \approx 260 \text{ W m}^{-2}$$

Refined line-by-line calculations:  $F_{\text{OLR}}^{\text{max}} \approx 280\text{--}310 \text{ W m}^{-2}$  (Kasting 1988; Goldblatt et al. 2013).

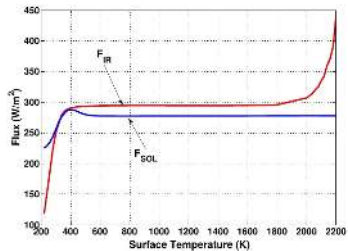
As  $T_s$  rises, the photosphere migrates upward but its temperature stays clamped: thermal flux saturates.

# Numerical Verification: Goldblatt 2013

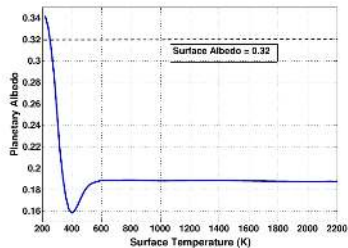


Credit: Goldblatt et al. (2013)

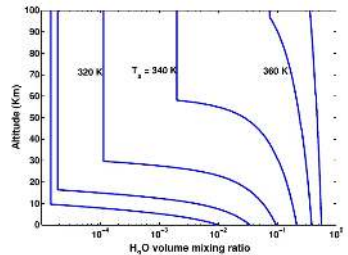
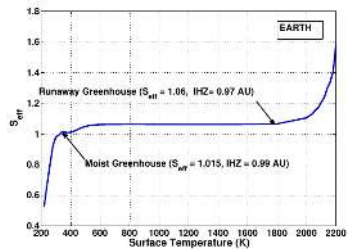
- Modelled OLR spectra at  $T_s = 280, 310, 340, 370, 400$  K
- Spectra collapse onto a common asymptote
- Asymptote integrates to  $\sim 282 \text{ W m}^{-2}$
- Validates the Simpson–Nakajima sketch quantitatively



(a)



(b)



## Reading the Goldblatt 2013 Panels

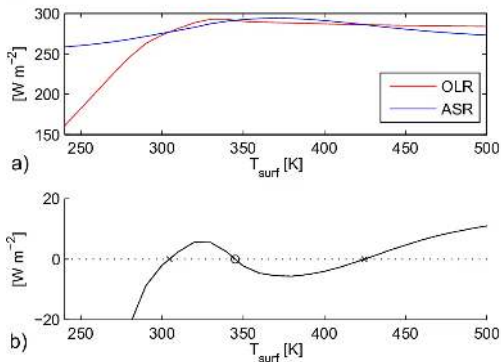
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- Top:  $F_{\text{OLR}}$  vs.  $T_s$  for varying  $\text{CO}_2$
- Middle: ASR vs.  $T_s$ , including water-cloud feedback
- Bottom:  $F_{\text{OLR}} - F_{\text{ASR}}$  as the steady-state condition
- Multiple equilibria (intersections) for some stellar fluxes: bistability
- Robust across reasonable choices of cloud + surface treatment

The Simpson–Nakajima limit is a thermodynamic feature, not a model artefact.



# Multiple Equilibria and Hysteresis



Credit: Wordsworth & Pierrehumbert (2013)

- Three intersections: cool-stable, warm-unstable, hot-stable
- For a range of stellar fluxes, two stable solutions coexist
- Once on the hot branch, no return to cool without removing  $\text{H}_2\text{O}$
- This is the source of the runaway hysteresis

## Step 4: Habitable Zone Implications

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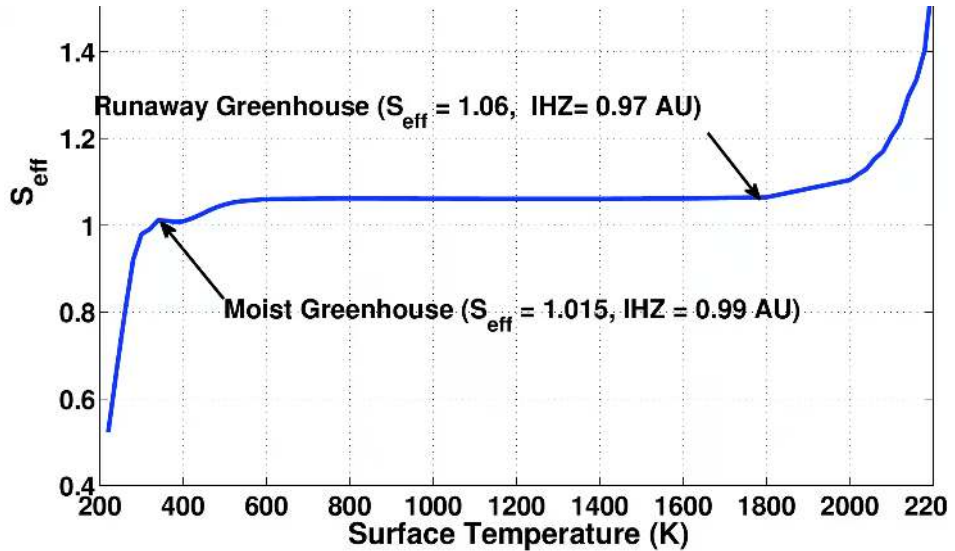
Compare to absorbed stellar flux at planetary orbit:

$$F_{\text{abs}} = \frac{S_{\odot}}{4a^2} (1 - A)$$

- Earth ( $a = 1 \text{ AU}$ ,  $A = 0.30$ ):  $F_{\text{abs}} = 240 \text{ W m}^{-2}$ , **below**  $F_{\text{OLR}}^{\text{max}}$  by  $\sim 40\text{--}70 \text{ W m}^{-2}$
- Venus ( $a = 0.723 \text{ AU}$ , Earth-like  $A$ ):  $F_{\text{abs}} \approx 460 \text{ W m}^{-2}$ , **far above**  $F_{\text{OLR}}^{\text{max}}$

Setting  $F_{\text{abs}} = F_{\text{OLR}}^{\text{max}}$  gives the inner HZ edge:

$$a_{\text{IHZ}} \approx 0.95\text{--}0.99 \text{ AU} \quad (\text{present Sun})$$



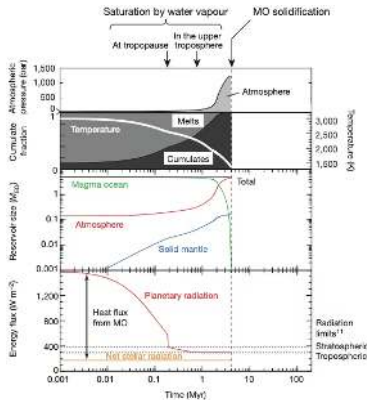
## Reading the Kopparapu HZ Plot

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- Effective stellar flux  $S_{\text{eff}}$  on the x-axis, surface  $T$  on the y-axis
- **Runaway greenhouse** limit:  $S_{\text{eff}} = 1.06$  ( $a = 0.97$  AU around present Sun)
- **Moist greenhouse** limit:  $S_{\text{eff}} = 1.015$  (0.99 AU)
- Both limits define the inner edge of the classical habitable zone
- Venus at  $a = 0.723$  AU corresponds to  $S_{\text{eff}} \approx 1.91$ , well inside the limit

**This figure is the canonical reference for the inner edge of the habitable zone.**

# Magma-Ocean Evolution: Type-I vs Type-II

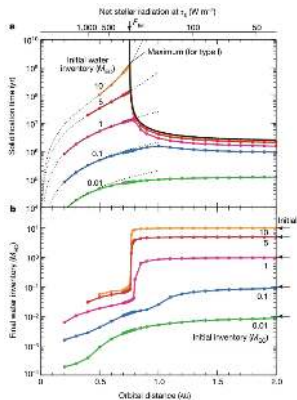


Credit: Hamano, Abe & Genda (2013)

Type-I planets (outside critical orbit):

- Magma ocean cools rapidly (< 10 Myr)
- Steam atmosphere condenses
- Liquid surface ocean forms
- Earth was Type-I

# The Type-I / Type-II Phase Diagram

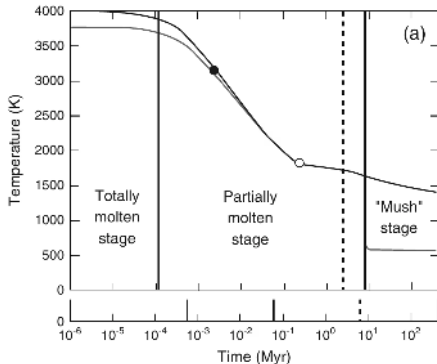


Credit: Hamano et al. (2013)

Critical orbital distance  $a_{cr} \approx 0.77$  AU:

- Inside  $a_{cr}$ : **Type-II**, magma ocean persists > 100 Myr, water lost to space
- Outside  $a_{cr}$ : **Type-I**, ocean condenses, water retained
- Venus straddles or sits inside  $a_{cr}$

# Magma-Ocean Atmosphere Coupling



Credit: Lebrun et al. (2013)

Coupled magma-ocean + atmosphere model:

- Solidification time vs. orbital distance
- For  $a < 0.66$  AU around a Sun-like star: indefinite delay
- Establishes the timing baseline for Venus' steam-atmosphere phase

## Why the Simpson-Nakajima Limit is Robust

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The limit depends only on:

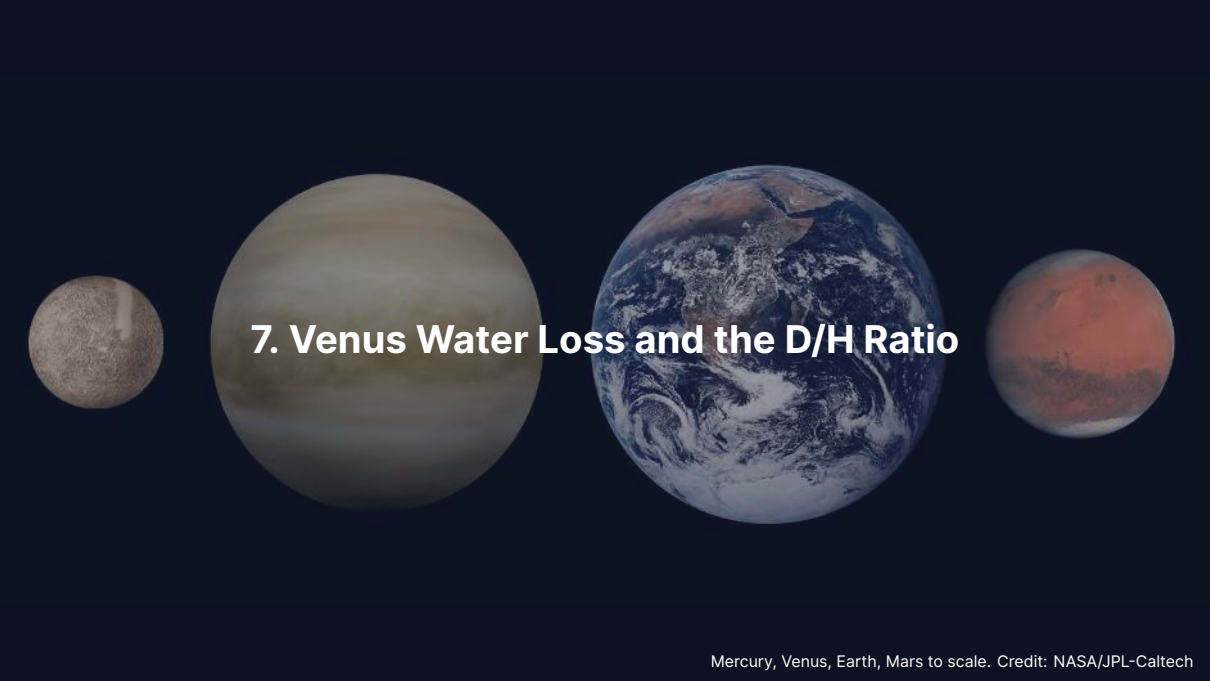
- Water phase diagram (Clausius–Clapeyron)
- Elementary radiative transfer ( $\tau \sim 1$  photosphere)

Insensitive to:

- Atmospheric composition (so long as water is abundant)
- Surface temperature (by construction)
- Detailed cloud microphysics (corrections at the 5–10% level)

Once  $F_{\text{abs}} > F_{\text{OLR}}^{\text{max}}$ , no steady state with surface liquid water. The transition is one-way: only photolysis + H escape returns the planet to a dry state, irreversibly.



The image displays four celestial bodies in a horizontal row against a dark, starry background. From left to right, they are Mercury, Venus, Earth, and Mars. Mercury is the smallest, appearing as a grey, cratered sphere. Venus is a large, pale yellowish-brown sphere with a hazy atmosphere. Earth is a large blue and white sphere showing continents and cloud patterns. Mars is a reddish-orange sphere with darker patches. The text '7. Venus Water Loss and the D/H Ratio' is centered over the Venus and Earth planets.

## 7. Venus Water Loss and the D/H Ratio

Mercury, Venus, Earth, Mars to scale. Credit: NASA/JPL-Caltech

## When Did Venus Lose Its Water?

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Two end-member scenarios, both physically plausible:

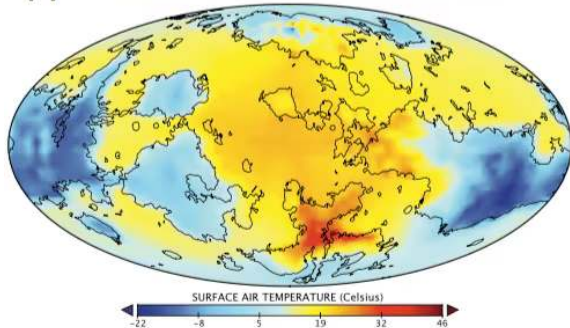
- **Early loss:** runaway from the magma-ocean stage; never had condensed oceans (Hamano Type-II)
- **Late loss:** temperate Earth-like surface for ~ 1 Gyr, runaway triggered by faint-young-Sun brightening (Way et al. 2016)

Distinguishing observation needed: surface mineralogy + atmospheric isotopes.

None of the present datasets is conclusive.

# Was Venus Ever Habitable? The Late-Loss Scenario

(a) Paleo Venus Surface Air Temperature (2.9Gya solar spectrum)



Credit: Way et al. (2016)

- GCM simulations of paleo-Venus with shallow ocean + slow rotation
- Surface  $T < 290$  K can persist for  $\sim 1$  Gyr
- Cloud feedback from slow rotation reflects flux back to space
- A habitable Venus is dynamically possible, not required

## The Loss Mechanism: Photolysis + Escape

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Sequential loss of water in the upper atmosphere:

1.  $\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{OH}$  (UV photolysis)
2. H escapes via Jeans + non-thermal channels (light, fast)
3. O recombines, slowly oxidises crust or escapes via solar-wind pickup

Mass balance: ~1 ocean equivalent of water can be lost over Gyr through this channel.

Without a magnetic field, the solar wind directly accesses the upper atmosphere  $\Rightarrow$  enhanced  $\text{O}^+$  pickup.

# The D/H Ratio: Smoking-Gun Evidence

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Measured atmospheric D/H ratios:

| Body                         | D/H ( $\times 10^{-4}$ ) |
|------------------------------|--------------------------|
| Protosolar (H <sub>2</sub> ) | ~ 0.20                   |
| Earth (oceans)               | 1.56                     |
| Comets (Halley, 67P)         | 1.5–5                    |
| <b>Venus</b>                 | ~ 240                    |
| Mars                         | ~ 8                      |

Source: Donahue et al. (1982); de Bergh et al. (1991)

- Venus' D/H is ~ 150× **Earth's**
- Lighter <sup>1</sup>H escapes more easily than heavier <sup>2</sup>H
- D enrichment ⇒ massive past H loss
- Initial water ≥ 1% of Earth's ocean mass

## Bounding the Initial Water Inventory

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Rayleigh fractionation in escape: residual water becomes D-enriched as H is lost.

Crude estimate (Jeans-like fractionation factor,  $\alpha \approx 0.5$ ):

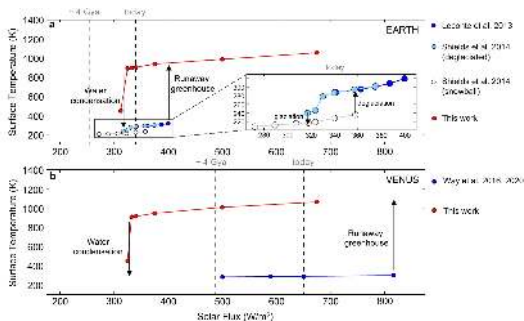
$$\frac{(D/H)_{\text{now}}}{(D/H)_{\text{init}}} = f^{\alpha-1} \approx f^{-0.5}$$

With observed enrichment factor  $\sim 150$ :

$$f \approx 150^{-2} \approx 4 \times 10^{-5}$$

Implies an initial water inventory  $\sim 25,000\times$  today's tiny atmospheric reservoir. Translates to several hundred metres of global equivalent layer (regime-dependent).

# Bistability and Hysteresis Loop



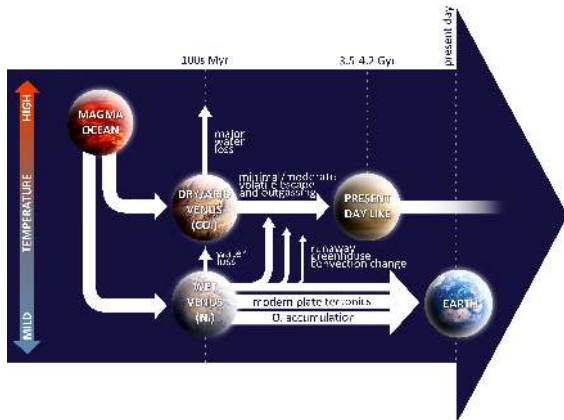
Credit: Turbet et al. (2021)

For some stellar fluxes, two stable states coexist:

- Wet: ocean + thin atmosphere
- Dry: hot, thick CO<sub>2</sub> atmosphere

Crossing the threshold is one-way.  
The dry state has no thermodynamic path back.

# Dry-Venus vs Wet-Venus Pathways



Credit: Gillmann et al. (2022)

Coupled-model branching:

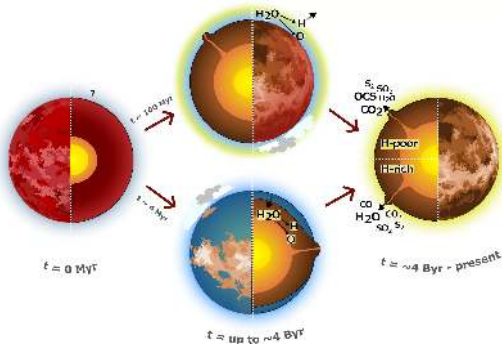
- Initial water inventory + outgassing rate determines branch
- Wet branch: like Earth, ocean condenses
- Dry branch: like Venus today
- Initial conditions decide everything



# Constraining the Trajectory

Statistical bracketing of Venus' history:

- Combines D/H, atmospheric noble gases, surface mineralogy
- Late-loss scenarios more probable in Bayesian framework
- Still room for early loss if early outgassing was anomalously low
- DAVINCI noble-gas measurements will narrow these constraints



Credit: Constantinou et al. (2024)



## 8. Why Did Earth and Venus Diverge?

## Carbonate–Silicate Cycle: Failure Mode

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On Earth, the carbon cycle keeps  $\text{CO}_2$  in the atmosphere ~ 280–1000 ppm via:

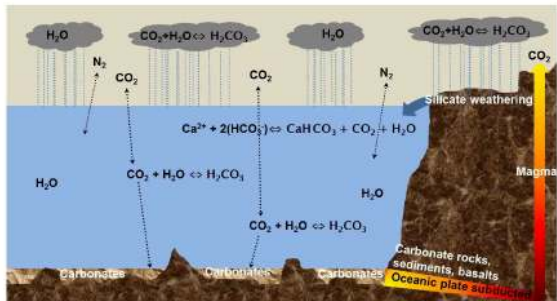
- Volcanic outgassing (source)
- Silicate weathering on continents (sink,  $T$ -dependent)
- Carbonate subduction + decarbonation (return loop)

On Venus this cycle has **no functional sink**:

- No liquid water  $\Rightarrow$  no aqueous weathering
- No subduction  $\Rightarrow$  no return of crustal carbonate to the mantle
- Volcanic outgassing accumulates indefinitely

Result:  $\text{CO}_2$  partial pressure builds to the 90-bar ambient observed today.

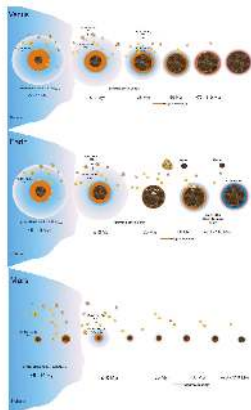
# The Carbonate-Silicate Schematic



Credit: Lammer et al. (2018)

- Earth: closed carbon loop with subduction return
- Mars: open loop (carbonates accumulate, no return without active tectonics)
- Venus: source-only (no aqueous chemistry, no subduction)

# Initial Volatile Inventories



Credit: Lammer et al. (2018)

- Comparative accretion of Venus, Earth, Mars
- Initial water + carbon inventories overlap substantially
- Divergence is not from initial conditions alone
- Early evolution decides which branch each planet takes

# Why Did the Twins Diverge?

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No single cause; several contributing factors:

1. **Distance from the Sun** ( $1.91\times$  Earth's flux)  $\Rightarrow$  inside the runaway-greenhouse limit
2. **No active dynamo**  $\Rightarrow$  no shielding from solar wind
3. **Slow rotation**  $\Rightarrow$  less Coriolis-driven cloud organisation
4. **Possible early magma-ocean phase**: Hamano Type-II, volatiles never fully condensed
5. **Stagnant lid tectonics**  $\Rightarrow$  no carbon-cycle return loop

All five may have contributed; their relative weight is an active research question.

## Implications for Habitability

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Venus matters far beyond Venus:

- Defines the inner edge of the classical habitable zone for any star
- Provides a **negative example** of what happens when key feedbacks fail
- Stagnant-lid + runaway combination may be the default for rocky planets, with Earth as the lucky exception
- Tidally-locked rocky planets near M-dwarfs are pre-conditioned to Venus-like outcomes

Venus is the calibration target for understanding which combinations of orbital, geophysical, and tectonic parameters allow long-term liquid water.



## 9. Recent Advances and Future Missions



Three Venus-bound missions over Magellan radar mosaic. Credit: Widemann et al. (2023)





DAVINCI (NASA), VERITAS (NASA), EnVision (ESA). Widemann et al. (2023)

## Three Missions to Venus, This Decade

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- **DAVINCI** (NASA, ~ 2030): atmospheric descent probe + flyby imaging
- **VERITAS** (NASA, ~ 2031): radar interferometry, surface composition mapping
- **EnVision** (ESA, ~ 2031): radar + subsurface sounder + atmospheric chemistry

Combined, they will measure: noble-gas isotopes (DAVINCI), high-resolution topography (VERITAS), subsurface structure + tidal Love number  $k_2$  (EnVision).

After a ~ 30-year gap in dedicated Venus missions, three at once.

## DAVINCI: A Descending Atmospheric Laboratory

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- Launch: nominally 2029–2030
- Hour-long parachute descent through the full atmosphere
- Mass spectrometer: noble gases + isotopes (Ar, Kr, Xe ratios)
- Tunable laser spectrometer: D/H,  $^{18}\text{O}/^{16}\text{O}$  at multiple altitudes
- Imaging: descent images of tessera highlands at < 100 m resolution

**Will measure noble-gas isotopes that distinguish early-loss vs. late-loss water scenarios for the first time.**

## VERITAS: Radar at 30 m Resolution

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- Synthetic aperture radar: ~ 30 m global topography (Magellan: ~ 100 m)
- InSAR: detect cm-scale surface deformation between flybys
- Near-IR emissivity: surface mineralogy through the atmosphere
- Gravity science: high-resolution gravity map,  $C/MR^2$  measurement

**Will measure  $C/MR^2$  for the first time, settling the core-size question.**

## EnVision: Surface, Subsurface, Atmosphere

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- ESA M5 mission, launch ~ 2031
- Synthetic aperture radar (VenSAR): selected high-resolution targets
- **Subsurface sounder**: layered structure of plains and tessera to ~ 1 km depth
- Spectrometers: mineralogy + atmospheric chemistry
- Radio science: gravity tides,  $k_2$  Love number  $\Rightarrow$  core liquidity

$k_2$  measurement will distinguish a liquid from solid core, addressing the missing-dynamo question.

## Open Questions for the 2030s

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1. Did Venus ever have surface oceans? For how long?
2. What is the present rate of volcanism?
3. Is the core liquid? Why no dynamo?
4. Does the atmosphere host disequilibrium chemistry of biological origin?
5. Does episodic-overtectonics operate, and on what timescale?

Venus is moving from a puzzle with no data to a puzzle with the right data on the way.

## Summary

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1. Earth and Venus have nearly identical bulk parameters
2. Their evolution diverged by  $\sim 500$  K in surface temperature,  $90\times$  in atmospheric mass, all surface water vs. none
3. Earth: plate tectonics + magnetic dynamo + biosphere + carbon cycle, all mutually reinforcing
4. Venus: stagnant lid + no dynamo + runaway greenhouse, all mutually reinforcing
5. The Simpson–Nakajima limit ( $\sim 280 \text{ W m}^{-2}$ ) is the thermodynamic boundary defining the inner edge of the habitable zone
6. Venus' D/H ratio of  $\sim 150\times$  Earth's is direct evidence of past water loss
7. DAVINCI, VERITAS, and EnVision will deliver the data to settle when, why, and how the divergence happened



# Questions?

Next: **Lecture 10. Rocky Planets II: Mercury & Mars**

Lecture notes: [formingworlds.github.io/IntroductionToPlanetaryScience](https://formingworlds.github.io/IntroductionToPlanetaryScience)

